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Inventor(s)

IIO; Atsuo et al.

FUEL CELL SYSTEM

Abstract

The fuel cell system includes a fuel cell, a fuel gas tank, a gas flow path, a pressure reducing valve, a pressure sensor for acquiring the pressure of the gas on the side of the fuel cell with respect to the pressure reducing valve, an operation storage unit for storing the length of the deactivation period of the fuel cell system, and a control unit. The control unit performs start control for starting the operation of the fuel cell when the first start condition including that the pressure acquired by the pressure sensor is smaller than the first pressure threshold is satisfied, and performs start control when the second start condition including that the pressure acquired by the pressure sensor is larger than the first pressure threshold and the length of the pause period stored in the operation storage unit is larger than the pause threshold is satisfied.

Inventors: IIO; Atsuo (Nagoya-shi, JP), YOKOTA; Keisuke (Nisshin-shi, JP), YAMAMOTO; Kazuo (Toyota-shi, JP), SHINODA; Morio (Anjo-shi, JP)

Applicant: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

Family ID: 96856742

Assignee: TOYOTA JIDOSHA KABUSHIKI KAISHA (Toyota-shi, JP)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-031216 filed on Mar. 1, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a fuel cell system.

2. Description of Related Art

[0003] Conventionally, there is a fuel cell system including a hydrogen tank, a shut-off valve, a pressure reducing valve, and a pressure regulating valve in a path to supply hydrogen to a fuel cell stack. The high-pressure hydrogen gas supplied from the hydrogen tank passes through the pressure reducing valve and the pressure regulating valve, is adjusted to an appropriate pressure and amount, and is supplied to the fuel cell stack. When the fuel cell system is not in operation, the pressure reducing valve and the pressure regulating valve are closed. There is a technique in which an abnormality is determined in such a fuel cell system when the pressure in the flow path between the pressure reducing valve and the pressure regulating valve is more than a threshold when an instruction to start operation is provided.

SUMMARY

[0004] When the fuel cell system is not operated for a long period, the pressure in the flow path between the pressure reducing valve and the pressure regulating valve may become more than the threshold because of the hydrogen gas that has passed through a seal portion of the pressure reducing valve during the period. In such a case, an abnormality is determined and the operation of the fuel cell system is restricted, even when there is no abnormality in the pressure reducing valve provided in the path to supply hydrogen.

[0005] The present disclosure can be implemented in the following aspects.

[0006] (1) An aspect of the present disclosure provides a fuel cell system. This fuel cell system includes: [0007] a fuel cell; [0008] a fuel gas tank that holds a fuel gas; [0009] a gas flow path that connects the fuel cell and the fuel gas tank to allow the fuel gas to flow through the gas flow path; [0010] a pressure reducing valve provided in the gas flow path, capable of allowing the fuel gas supplied from the fuel gas tank to flow through the pressure reducing valve with a pressure of the fuel gas reduced, and capable of shutting off the fuel gas supplied from the fuel gas tank; [0011] a pressure sensor capable of acquiring a pressure in the gas flow path on a side of the fuel cell with respect to the pressure reducing valve; [0012] an operation storage unit that stores a length of a quiescent period of the fuel cell system; and [0013] a control unit that controls the fuel cell system. [0014] The control unit performs start control for starting operation of the fuel cell when an instruction to start the operation of the fuel cell is received, and [0015] when a first start condition is met, the first start condition including the pressure acquired by the pressure sensor being less than a first pressure threshold determined in advance; and the control unit performs the start control when a second start condition is met, the second start condition including the pressure acquired by the pressure sensor being more than the first pressure threshold and the length of the quiescent period stored in the operation storage unit being more than a quiescent threshold determined in advance. [0016] According to this aspect, it is possible to avoid a situation in which the start

control is not performed when the pressure in the gas flow path on the fuel cell side with respect to the pressure reducing valve becomes more than the first pressure threshold since the quiescent period in which the operation of the fuel cell is not performed is more than the quiescent threshold, even if there is no abnormality in the function of the pressure reducing valve. That is, the start control for starting the operation of the fuel cell can be performed as when the pressure in the gas flow path on the fuel cell side with respect to the pressure reducing valve is less than the first pressure threshold. [0017] (2) The fuel cell system according to the above aspect may further include [0018] an adjusting unit provided in the gas flow path on a side of the fuel cell with respect to the pressure reducing valve to receive the fuel gas and supply the fuel gas to the fuel cell at a specified opening degree, and [0019] a power supply storage unit that stores a shutoff of supply of power to the operation storage unit; [0020] the pressure sensor may acquire a pressure in the gas flow path between the pressure reducing valve and the adjusting unit; and the control unit may perform the start control when an instruction to start the operation of the fuel cell is received and when a third start condition is met, the third start condition including [0021] (i) the pressure acquired by the pressure sensor being more than the first pressure threshold, [0022] (ii) a shutoff of supply of power to the operation storage unit being stored, and [0023] (iii) the pressure acquired by the pressure sensor becoming less than a second pressure threshold determined in advance after the adjusting unit is caused to supply the fuel gas toward the fuel cell. [0024] According to this aspect, it is possible to avoid a situation in which the start control is not performed, even if there is no abnormality in the function of the pressure reducing valve, when the pressure in the gas flow path on the fuel cell side with respect to the pressure reducing valve becomes more than the first pressure threshold since the length of the quiescent period in which the operation of the fuel cell is not performed is more than the quiescent threshold, and when the length of the quiescent period is not stored in the operation storage unit since the supply of power to the operation storage unit is shut off. In that event, it is possible to avoid a situation in which the start control is performed even if there is an abnormality in the function of the pressure reducing valve, since the start control is performed on condition that the pressure acquired by the pressure sensor becomes less than the second pressure threshold after the adjusting unit is caused to supply the fuel gas toward the fuel cell. [0025] (3) The fuel cell system according to the above aspect may further include [0026] a main stop valve provided in the gas flow path on a side of the fuel gas tank with respect to the pressure reducing valve to allow or shut off a flow of the fuel gas; and [0027] the pressure acquired by the pressure sensor becoming less than the second pressure threshold after the adjusting unit is caused to supply the fuel gas toward the fuel cell may include the pressure acquired by the pressure sensor becoming less than the second pressure threshold within a time threshold determined in advance after the adjusting unit is caused to supply the fuel gas toward the fuel cell with the main stop valve in a closed state. [0028] According to this aspect, a reduction in the pressure on the downstream side of the pressure reducing valve is checked with the main stop valve in the closed state. Therefore, it is possible to reduce the possibility that the fuel gas is ejected to the outside with the adjusting unit exposed to the high-pressure fuel gas to be broken, or with a relief mechanism provided in the adjusting unit opened, when the pressure reducing valve has failed. [0029] (4) Another aspect of the present disclosure provides a fuel cell system. [0030] This fuel cell system includes: [0031] a fuel cell; [0032] a fuel gas tank that holds a fuel gas; [0033] a gas flow path that connects the fuel cell and the fuel gas tank to allow the fuel gas to flow through the gas flow path; [0034] a pressure reducing valve provided in the gas flow path, capable of allowing the fuel gas supplied from the fuel gas tank to flow through the pressure reducing valve with a pressure of the fuel gas reduced, and capable of shutting off the fuel gas supplied from the fuel gas tank; [0035] an adjusting unit provided in the gas flow path on a side of the fuel cell with respect to the pressure reducing valve to receive the fuel gas and supply the fuel gas to the fuel cell at a specified opening degree; [0036] a pressure sensor capable of acquiring a pressure in the gas flow path between the pressure reducing valve and the adjusting unit; and a control unit that controls the fuel cell system.

The control unit performs start control for starting operation of the fuel cell when an instruction to start the operation of the fuel cell is received, and [0037] when a start condition is met, the start condition including the pressure acquired by the pressure sensor being less than a first pressure threshold determined in advance; and [0038] the control unit performs the start control when an additional start condition is met, the additional start condition including [0039] (i) the pressure acquired by the pressure sensor being more than the first pressure threshold, and [0040] (ii) the pressure acquired by the pressure sensor becoming less than a second pressure threshold determined in advance after the adjusting unit is caused to supply the fuel gas toward the fuel cell. [0041] According to this aspect, it is possible to avoid a situation in which the start control is not performed, even if there is no abnormality in the function of the pressure reducing valve, when the pressure in the gas flow path on the fuel cell side with respect to the pressure reducing valve becomes more than the first pressure threshold since the quiescent period in which the operation of the fuel cell is not performed is long. In that event, it is possible to avoid a situation in which the start control is performed even if there is an abnormality in the function of the pressure reducing valve, since the start control is performed on condition that the pressure acquired by the pressure sensor becomes less than the second pressure threshold after the adjusting unit is caused to supply the fuel gas toward the fuel cell. [0042] The present disclosure can be implemented in various forms other than the fuel cell system. For example, the present disclosure can be implemented in the form of a method of manufacturing a fuel cell system, a method of controlling a fuel cell system, a computer program that implements the control method, a non-transitory storage medium storing the computer program, etc.

BRIEF DESCRIPTION OF THE DRAWINGS

[0043] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0044] FIG. 1 is a schematic diagram showing a schematic configuration of a fuel cell system **100** according to the present embodiment;

[0045] FIG. 2 is a flow chart showing a process executed when an instruction Is indicating that the operation of the fuel cell **20** should be started is received;

[0046] FIG. 3 is a flow chart showing a process in the second embodiment executed when an instruction Is indicating that the operation of the fuel cell **20** should be started is received;

[0047] FIG. 4 is a flow chart showing a process in the third embodiment executed when an instruction Is indicating that the operation of the fuel cell **20** should be started is received; and

[0048] FIG. 5 is a flow chart showing a process of the third embodiment executed when an instruction Is indicating that the operation of the fuel cell **20** should be started is received.

Description

DETAILED DESCRIPTION OF EMBODIMENTS

A. First Embodiment

[0049] FIG. 1 is a schematic diagram illustrating a schematic configuration of a fuel cell system **100** according to the present embodiment. The fuel cell system **100** is mounted on a vehicle. The fuel cell system **100** supplies electric power to a load **300** mounted on the vehicle.

[0050] The vehicle includes a fuel cell system **100**, a secondary battery **200**, and a load **300**. The load **300** functions by being supplied with electric power from the fuel cell system **100** and the secondary battery **200**. The load **300** includes a drive motor for moving the vehicle, an air conditioning facility for adjusting a temperature in the vehicle cabin, and the like.

[0051] The fuel cell system **100** includes a fuel cell **20**, a cathode gas supply/discharge system **30**, an anode gas supply/discharge system **50**, a DC/DC converter **70**, and a control device **90**.

[0052] In the fuel cell **20**, air as a cathode gas is supplied from the cathode gas supply/discharge system **30**, and hydrogen gas as an anode gas is supplied from the anode gas supply/discharge system **50** to generate electric power (see the lower center portion in FIG. **1**). The fuel cell **20** is a polymer electrolyte fuel cell. The fuel cell **20** has a stack structure including a plurality of stacked single cells **21**. The single cell **21** includes an anode **22** and a cathode **23**. The single cell **21** generates electric power by supplying hydrogen gas to the anode **22** and air to the cathode **23**. The single cell **21** constitutes the smallest unit of power generation. In FIG. **1**, the anode **22** and the cathode **23** of each unit cell **21** are collectively shown for ease of understanding of the technology.

[0053] The cathode gas supply/discharge system **30** supplies the cathode gas to the fuel cell **20**, and discharges the cathode off-gas discharged from the fuel cell **20** to the outside of the fuel cell system **100** (see the left part of the middle stage in FIG. **1**). The cathode gas supply/discharge system **30** includes a cathode gas supply flow path **31**, an air flow meter **32**, a compressor **33**, a cathode pressure sensor **34**, a flow regulating valve **36**, a bypass flow path **41**, a flow dividing valve **42**, and a cathode off-gas discharge flow path **45**.

[0054] The cathode gas supply flow path **31** is a flow path for supplying the air taken in from the outside of the fuel cell system **100** to the fuel cell **20** as the cathode gas (see the left part of the middle stage in FIG. **1**). In the cathode gas supply flow path **31**, an air flow meter **32**, a compressor **33**, a cathode pressure sensor **34**, and a flow regulating valve **36i** are provided in that order.

[0055] The air flow meter **32** detects the flow rate of the cathode gas taken into the cathode gas supply flow path **31**. The compressor **33** compresses the cathode gas supplied by the cathode gas supply flow path **31** and supplies the compressed cathode gas to the downstream side. The cathode pressure sensor **34** detects the pressure in the cathode gas supply flow path **31** on the fuel cell **20** side with respect to the compressor **33**. The flow regulating valve **36i** regulates the flow rate of the cathode gas flowing through the cathode gas supply flow path **31**.

[0056] The cathode off-gas discharge flow path **45** is a flow path for discharging the cathode off-gas discharged from the fuel cell **20** to the outside of the fuel cell system **100** (see the lower left portion in FIG. **1**). A flow regulating valve **360** is provided in the cathode off-gas discharge flow path **45**. The flow regulating valve **360** regulates the flow rate of the cathode off-gas flowing through the cathode off-gas discharge flow path **45**. In this specification, the flow regulating valve **36i** and the flow regulating valve **360** are collectively referred to as a “flow regulating valve **36**”.

[0057] The bypass flow path **41** is a flow path connecting the cathode gas supply flow path **31** and the cathode off-gas discharge flow path **45** (see the left part of the middle stage in FIG. **1**). A flow dividing valve **42** is provided in the bypass flow path **41**. The flow dividing valve **42** can adjust the flow rate of the cathode gas flowing through the bypass flow path **41**, and can further block the flow of the cathode gas in the bypass flow path **41**. The flow of the cathode gas supplied from the cathode gas supply flow path **31** is divided into a flow toward the fuel cell **20** and a flow toward the bypass flow path **41** by the flow dividing valve **42**.

[0058] The anode gas supply/discharge system **50** supplies the anode gas to the fuel cell **20**, and discharges the anode off-gas discharged from the fuel cell **20** to the outside of the fuel cell system **100** (see the lower right part in FIG. **1**). The anode gas supply/discharge system **50** includes an anode gas tank **51**, an anode gas supply passage **52**, a main stop valve **54**, a pressure reducing valve **55**, an adjusting unit **56**, and pressure sensors **53**, **57**, and **58**. The anode gas supply/discharge system **50** further includes an anode gas discharge path **61**, a gas-liquid separator **62**, a circulation flow path **63**, an anode gas pump **64**, and an exhaust drain valve **65**.

[0059] The anode gas tank **51** holds the anode gas (see the right part of the middle stage in FIG. **1**). More specifically, the anode gas tank **51** stores hydrogen gas having a pressure higher than that of the hydrogen gas supplied to the fuel cell **20**.

[0060] The anode gas supply passage **52** is a flow passage that connects the fuel cell **20** and the anode gas tank **51** and allows the anode gas to flow therethrough (see the right portion of the middle stage in FIG. **1**). In the anode gas supply passage **52**, a pressure sensor **53**, a main stop

valve **54**, a pressure reducing valve **55**, a pressure sensor **57**, an adjusting unit **56**, and a pressure sensor **58** are provided in that order.

[0061] The main stop valve **54** is provided in the anode gas supply passage **52** at a position on the side of the anode gas tank **51** with respect to the pressure reducing valve **55**. The main stop valve **54** allows or blocks the flow of the anode gas.

[0062] The pressure reducing valve **55** can lower and flow the pressure of the anode gas supplied from the anode gas tank **51**, and can shut off the anode gas supplied from the anode gas tank **51**.

[0063] The adjusting unit **56** is provided in the anode gas supply passage **52** on the side of the fuel cell **20** with respect to the pressure reducing valve **55**. The adjusting unit **56** may take various degrees of opening. The larger the opening degree, the larger the flow rate of the anode gas flowing downstream from the adjusting unit **56**. The adjusting unit **56** receives the anode gas and supplies the anode gas to the fuel cell **20** at a specified opening degree. Specifically, the adjusting unit **56** is a linear solenoid valve.

[0064] The pressure sensors **53**, **57**, and **58** acquire the pressure in the anode gas supply passage **52**. The pressure sensor **53** acquires the pressure of the anode gas supplied from the anode gas tank **51**. Specifically, the pressure sensor **53** acquires the pressure in the anode gas supply passage **52** on the side of the anode gas tank **51** with respect to the main stop valve **54**. The pressure sensor **57** can acquire the pressure in the anode gas supply passage **52** on the side of the fuel cell **20** with respect to the pressure reducing valve **55**. More specifically, the pressure sensor **57** acquires the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56**. The pressure sensor **58** acquires the pressure in the anode gas supply passage **52** on the fuel cell **20** side with respect to the connection portion with the circulation flow path **63**.

[0065] The anode gas discharge path **61** is a flow path that connects the fuel cell **20** and the gas-liquid separator **62** and supplies the anode off-gas discharged from the fuel cell **20** to the gas-liquid separator **62** (see the lower center portion in FIG. **1**). The anode off-gas discharged from the fuel cell **20** includes an anode gas, liquid water, and an impurity gas that are not used in the electrochemical reaction in the fuel cell **20**.

[0066] The gas-liquid separator **62** separates the liquid water from the anode off-gas containing the liquid water (see the lower middle portion in FIG. **1**). An exhaust drain valve **65** is connected to the gas-liquid separator **62**.

[0067] The exhaust drain valve **65** allows or blocks the discharge of liquid water and impurity gases from the gas-liquid separator **62**. The exhaust drain valve **65** is normally closed and opens in response to a command from the control device **90**. As a result, the liquid water and the impurity gas separated from the anode off-gas by the gas-liquid separator **62** are discharged to the outside of the fuel cell system **100** via the cathode off-gas discharge flow path **45**.

[0068] The circulation flow path **63** connects the portion of the anode gas supply passage **52** on the fuel cell **20** side to the adjusting unit **56** and the gas-liquid separator **62** (see the middle portion of FIG. **1**). The circulation flow path **63** is a flow path for supplying the anode off-gas from which the liquid water and the impurity gas are removed, which is supplied from the gas-liquid separator **62**, to the anode gas supply passage **52**. An anode gas pump **64** is provided in the circulation flow path **63**.

[0069] The anode gas pump **64** supplies the anode off-gas supplied from the gas-liquid separator **62** to the anode gas supply passage **52**. As a result, the anode off-gas supplied through the gas-liquid separator **62** and the anode gas supplied from the anode gas tank **51** are supplied to the fuel cell **20** through the anode gas supply passage **52**.

[0070] DC/DC converters **70** are disposed between electrically connected loads **300** and the fuel cell **20** (see the middle section of FIG. **1**). DC/DC converters **70** receive power from the fuel cell **20** and convert the voltage to a desired voltage. The power outputted by DC/DC converters **70** is supplied to the loads **300** via inverters.

[0071] In FIG. **1**, a fuel cell system **100** and a load **300** are shown as separate configurations for

ease of understanding of the technology. However, the load **300** includes auxiliary equipment such as the compressor **33** and the anode gas pump **64** of the fuel cell system **100** in addition to the drive motor and the air conditioning equipment of the vehicle.

[0072] The control device **90** controls the fuel cell system **100** (see the upper part of FIG. **1**). The control device **90** is a so-called ECU (Electronic Control Unit). The control device **90** includes a control unit **92** that is a CPU (Central Processing Unit), an operation storage unit **94**, a power supply storage unit **96**, an input unit **97**, and an output unit **98**.

[0073] The control unit **92** controls the fuel cell system **100**. Specifically, the control unit **92** controls the fuel cell system **100** by executing a program stored in the operation storage unit **94**. In addition to the sensors included in the fuel cell system **100** such as the cathode pressure sensor **34**, the pressure sensor **53**, and the pressure sensor **57**, a detection signal is input to the control unit **92** from a sensor group such as an accelerator operation amount sensor or a vehicle speed sensor of the vehicle. The control unit **92** controls each unit by outputting a drive signal to each unit related to power generation of the fuel cell **20**, including the main stop valve **54**, the pressure reducing valve **55**, the adjusting unit **56**, the anode gas pump **64**, and the exhaust drain valve **65**. For example, the control unit **92** instructs the adjusting unit **56** to indicate the opening degree.

[0074] The control unit **92** records the length T_{ia} of the deactivation period of the fuel cell system **100** in the operation storage unit **94**. The rest period is a period in which the anode gas and the cathode gas are not supplied to the fuel cell **20** and power generation is not performed in the fuel cell **20**. Specifically, the control unit **92** records, in the operation storage unit **94**, the length of the period from the input of the instruction to end the operation of the fuel cell system **100** via the input unit **97** to the input of the instruction to start the operation of the fuel cell system **100** via the input unit **97** as the length T_{ia} of the pause period. That is, when an instruction to end the operation of the fuel cell system **100** is input, the control unit **92** records the time at that time in the operation storage unit **94**, stops the supply of the anode gas and the cathode gas to the fuel cell **20**, and stops the power generation by the fuel cell **20**. After that, when an instruction to start the operation of the fuel cell system **100** is first input, the control unit **92** records the time at that time in the operation storage unit **94**, resumes the supply of the anode gas and the cathode gas to the fuel cell **20**, and resumes the power generation by the fuel cell **20**. Then, the control unit **92** records, in the operation storage unit **94**, the length of the period from the time at which the instruction to end the operation of the fuel cell system **100** is input to the time at which the instruction to start the operation of the fuel cell system **100** is input first, as the length T_{ia} of the pause period.

[0075] The control unit **92** records the interruption of the supply of electric power to the operation storage unit **94** in the power supply storage unit **96**. The interruption of the supply of electric power to the operation storage unit **94** occurs, for example, when the secondary battery **200** is removed from the vehicle and replaced. The function of the control unit **92** may be realized by software or by a hardware circuit.

[0076] The operation storage unit **94** stores a program for controlling the fuel cell system **100** and data used for controlling the fuel cell system **100**. The operation storage unit **94** stores, for example, the length T_{ia} of the last pause period among the pause periods of the fuel cell system **100**. Specifically, the operation storage unit **94** includes RAM (Random Access Memory) and ROM (Read Only Memory). The length T_{ia} of the deactivation period of the fuel cell system **100** is stored in RAM.

[0077] The power supply storage unit **96** stores a program for controlling the fuel cell system **100** and data used for controlling the fuel cell system **100**. The power supply storage unit **96** stores, for example, interruption of power supply to the operation storage unit **94**. Specifically, the power supply storage unit **96** includes a rewritable nonvolatile memory. Therefore, the history of interruption of the supply of power to the operation storage unit **94** stored in the power supply storage unit **96** is not lost even if the supply of power to the fuel cell system **100** is interrupted.

[0078] The input unit **97** receives an instruction related to the operation of the fuel cell system **100**

from the driver of the vehicle, and inputs the instruction to the control unit **92**. Specifically, the input unit **97** is a switch for receiving an instruction to start or end the operation of the fuel cell **20**, or a touch panel for instructing an operation mode of the fuel cell **20**.

[0079] The output unit **98** is controlled by the control unit **92** and outputs information related to the operation of the fuel cell system **100** to the driver of the vehicle. Specifically, the output unit **98** is a liquid crystal display.

[0080] FIG. **2** is a flow chart showing a process executed when an instruction *Is* indicating that the operation of the fuel cell **20** should be started is received (see the upper part of FIG. **1**). The processing illustrated in FIG. **2** is executed by the control unit **92**.

[0081] In step **110**, the control unit **92** acquires the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** from the pressure sensor **57** (see the right part of the middle stage in FIG. **1**). At this time, the main stop valve **54** and the pressure reducing valve **55** are closed.

[0082] In step **115**, the control unit **92** determines whether the pressure P_m acquired by the pressure sensor **57** is smaller than a predetermined first pressure threshold P_{th1} . If the pressure P_m is less than the first pressure threshold P_{th1} , the process proceeds to step **140**. If the pressure P_m is greater than or equal to the first pressure threshold P_{th1} , the process proceeds to step **120**.

[0083] In step **120**, the control unit **92** acquires the length T_{ia} of the pause period stored in the operation storage unit **94** (see the upper part of FIG. **1**).

[0084] In step **125**, the control unit **92** determines whether or not the length T_{ia} of the pause period stored in the operation storage unit **94** is greater than a predetermined pause threshold T_{thia} . If the length T_{ia} of the pause period is greater than the pause threshold T_{thia} , the process proceeds to step **130**. If the length T_{ia} of the pause period is less than or equal to the pause threshold T_{thia} , the process proceeds to step **150**.

[0085] In step **130**, the pressure abnormality detection process is masked. That is, the process of step **150** is not executed, and the process proceeds to step **140**.

[0086] In step **140**, the control unit **92** starts the normal start control. Specifically, valves such as the main stop valve **54** and the pressure reducing valve **55** are opened, and start control for starting the operation of the fuel cell **20** is performed (see the right part of the middle stage in FIG. **1**).

[0087] In the process of FIG. **2**, the process of step **140** is performed when Yes determination is performed in step **115** and when Yes determination is performed in step **125**. The case where Yes determination is performed in step **115** is a case where the first starting condition that the pressure P_m acquired by the pressure sensor **57** is smaller than the predetermined first pressure threshold P_{th1} is satisfied.

[0088] In step **150**, the control unit **92** executes a pressure abnormality detection process. Step **150** includes step **152,154**. In step **152**, the control unit **92** performs fail-safe processing. Specifically, the control unit **92** performs setting so that, even if an instruction *Is* indicating that the operation of the fuel cell **20** should be started is input via the input unit **97**, the power is turned off immediately after the power is turned on. This setting is stored in the non-volatile memory. In step **154**, the control unit **92** performs start control at the time of abnormality. Specifically, the control unit **92** closes the main stop valve **54** and stops the fuel cell system **100**.

[0089] For example, when the pressure reducing valve **55** cannot sufficiently shut off the anode gas due to the deterioration of the seal portion, the anode gas in the anode gas supply passage **52** between the pressure reducing valve **55** and the main stop valve **54** enters the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** through the pressure reducing valve **55** during the deactivation period of the fuel cell system **100**. Consequently, even in a short period in which the length T_{ia} of the pause period is equal to or less than the pause threshold T_{thia} , the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** becomes equal to or greater than the pause threshold T_{thia} (see **S115: No** and **S125: No** in FIG. **2**). In the present embodiment, in such a case, the pressure

abnormality detection process in step **150** is performed. Therefore, when there is an abnormality in the function of the pressure reducing valve **55**, it is possible to avoid the operation of the fuel cell system **100** contrary to the instruction from the driver.

[0090] On the other hand, even when the pressure reducing valve **55** is capable of sufficiently shutting off the anode gas, the anode gas in the anode gas supply passage **52** between the pressure reducing valve **55** and the main stop valve **54** leaks slightly from the pressure reducing valve **55** and enters the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56**. Therefore, when the deactivation period of the fuel cell system **100** is long, the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** may be equal to or higher than the deactivation threshold T_{thia} . Such a condition may occur, for example, when the owner of the vehicle replaces the pressure reducing valve **55** after the purchase of the vehicle.

[0091] In the present embodiment, when the first start condition including that the pressure P_m acquired by the pressure sensor **57** is smaller than the predetermined first pressure threshold P_{th1} is satisfied, start control for starting the operation of the fuel cell **20** is performed (see **S115: Yes** and **S140** in FIG. 2). Further, the start control is also performed when the second start condition including that the pressure P_m acquired by the pressure sensor **57** is larger than the first pressure threshold P_{th1} and the length T_{ia} of the pause period stored in the operation storage unit **94** is larger than the predetermined pause threshold T_{thia} is satisfied (see **S115: No**, **S125: Yes** and **S140** in FIG. 2).

[0092] Therefore, in the present embodiment, since the pause period in which the operation of the fuel cell **20** is not performed is larger than the pause threshold T_{thia} even though the function of the pressure reducing valve **55** is not abnormal, it is possible to avoid a situation in which the starting control is not performed when the pressure in the anode gas supply passage **52** on the fuel cell **20** side becomes larger than the first pressure threshold P_{th1} with respect to the pressure reducing valve **55**. That is, the starting control for starting the operation of the fuel cell **20** can be performed in the same manner as when the pressure in the anode gas supply passage **52** on the fuel cell **20** side is smaller than the first pressure threshold P_{th1} with respect to the pressure reducing valve **55**.

[0093] The anode gas in the present embodiment is also referred to as “fuel gas”. The anode gas tank **51** is also called a “fuel gas tank”. The anode gas supply passage **52** is also referred to as a “gas flow path”.

B. Second Embodiment

[0094] FIG. 3 is a flow chart showing a process in the second embodiment executed when an instruction I_s indicating that the operation of the fuel cell **20** should be started is received (see the upper part of FIG. 1). In the process of FIG. 3, the process of step **122,127** is executed instead of the process of step **120,125** of FIG. 2. The configuration and processing of the fuel cell system of the second embodiment are the same as those of the fuel cell system **100** of the first embodiment. In FIG. 3, the steps that perform the same processing as the steps illustrated in FIG. 2 are denoted by the same reference numerals.

[0095] In step **122**, the control unit **92** opens the main stop valve **54** and causes the adjusting unit **56** to supply the anode gas. At this time, the pressure reducing valve **55** is closed.

[0096] After a predetermined time elapses after the fuel gas is supplied to the adjusting unit **56**, the control unit **92** acquires the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** from the pressure sensor **57** (see the right part of the middle stage in FIG. 1).

[0097] In step **127**, the control unit **92** determines whether the pressure P_m acquired by the pressure sensor **57** is smaller than a predetermined second pressure threshold P_{th2} .

[0098] In the anode gas supply passage **52**, when the pressure reducing valve **55** is capable of sufficiently shutting off the anode gas, when the fuel gas is supplied to the adjusting unit **56**, the anode gas in the anode gas supply passage **52** between the pressure reducing valve **55** and the

adjusting unit **56** flows downstream. Therefore, the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** is greatly reduced. If so, a Yes determination is made at step **127**.

[0099] On the other hand, in the anode gas supply passage **52**, when the pressure reducing valve **55** cannot sufficiently shut off the anode gas, the anode gas between the pressure reducing valve **55** and the adjusting unit **56** flows downstream, while the anode gas of the anode gas tank **51** enters the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** via the main stop valve **54** and the pressure reducing valve **55**. Therefore, even if the fuel gas is supplied to the adjusting unit **56**, the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** does not decrease. If so, a No determination is made at step **127**.

[0100] In step **127**, if the pressure P_m is less than the second pressure threshold P_{th2} , the process proceeds to step **130**. If the pressure P_m is greater than or equal to the second pressure threshold P_{th2} , the process proceeds to step **150**.

[0101] In the second embodiment, when the start condition including that the pressure P_m acquired by the pressure sensor **57** is smaller than the predetermined first pressure threshold P_{th1} is satisfied, the start control for starting the operation of the fuel cell **20** is performed (see **S115**: Yes and **S140** in FIG. 3). In addition, [0102] (i) The pressure P_m acquired by the pressure sensor **57** is larger than the first pressure threshold P_{th1} (see **S115**: No of FIG. 3), and [0103] (ii) After the fuel gas is supplied to the adjusting unit **56** toward the fuel cell **20**, the pressure P_m acquired by the pressure sensor **57** becomes smaller than the predetermined second pressure threshold P_{th2} (see **S122** and **S127**: Yes in FIG. 3), [0104] If additional start criteria are met, including, start control is also performed (see **S140** in FIG. 3).

[0105] By performing such a process, since the length T_{ia} of the deactivation period in which the operation of the fuel cell **20** has not been performed is long, when the pressure in the anode gas supply passage **52** on the fuel cell **20** side becomes larger than the first pressure threshold P_{th1} with respect to the pressure reducing valve **55**, it is possible to avoid a situation in which the starting control is not performed even though the function of the pressure reducing valve **55** is not abnormal. At this time, since the pressure P_m acquired by the pressure sensor **57** is smaller than the second pressure threshold P_{th2} after the fuel gas is supplied to the adjusting unit **56** toward the fuel cell **20**, a situation in which the starting control is performed can be avoided even though the function of the pressure reducing valve **55** is abnormal (see **S127**: Yes in FIG. 3).

[0106] In a second embodiment, [0107] (i) The pressure P_m acquired by the pressure sensor **57** is larger than the first pressure threshold P_{th1} (see **S115**: No of FIG. 3), and [0108] (ii) After the fuel gas is supplied to the adjusting unit **56** toward the fuel cell **20**, the pressure P_m acquired by the pressure sensor **57** is not smaller than the first pressure threshold P_{th1} within a predetermined time (see **S127**: No of FIG. 3). [0109] When the additional stop condition is satisfied, the start control is not performed, and the pressure-abnormality detection process is executed (see **S150** in FIG. 3).

[0110] Therefore, when the pressure reducing valve **55** cannot sufficiently shut off the anode gas supplied from the anode gas tank **51**, it is possible to avoid the operation of the fuel cell system **100** contrary to the instruction from the driver.

C. Third Embodiment

[0111] FIG. 4 and FIG. 5 are flow charts showing processes in the third embodiment executed when an instruction I_s indicating that the operation of the fuel cell **20** should be started is received (see the upper part of FIG. 1). In the processing of FIGS. 4 and 5, the processing of steps **116** to **119** is executed instead of the processing of step **122** of FIG. 3. Further, the process branches from step **117**, and the process of step **120,125,132** is executed. The configuration and processing of the fuel cell system of the third embodiment are the same as those of the fuel cell system of the second embodiment. In FIGS. 4 and 5, steps that perform the same processing as the steps illustrated in FIGS. 2 and 3 are denoted by the same reference numerals.

[0112] In step **116**, the control unit **92** refers to the history of interruption of power supply to the operation storage unit **94** recorded in the power supply storage unit **96**.

[0113] In step **117**, the control unit **92** determines whether or not the interruption of the supply of power to the operation storage unit **94** is stored in the power supply storage unit **96**. If the interruption of the supply of power to the operation storage unit **94** is stored in the power supply storage unit **96**, the process proceeds to step **118**. If the interruption of the supply of power to the operation storage unit **94** is not stored in the power supply storage unit **96**, the process proceeds to step **120** (see A in FIG. 4 and the upper part of FIG. 5).

[0114] The processing of step **120,125** is the same as the processing of step **120,125** in the first embodiment. In step **125**, if the length T_{ia} of the pause period is less than or equal to the pause threshold T_{thia} , the process proceeds to step **150** (see B of FIG. 5 and the lower right part of FIG. 4). If the length T_{ia} of the pause period is greater than the pause threshold T_{thia} , the process proceeds to step **130c**. As can be distinguished from step **130** shown in FIG. 4, step **130** is shown as “step **130c**” in FIG. 5. The process in step **130c** is the same as step **130** in the first embodiment. After the step **130c**, the process proceeds to step **140** (see C in FIG. 5 and the lower part of FIG. 4).

[0115] That is, also in the third embodiment, when the second starting condition including that the pressure P_m acquired by the pressure sensor **57** is larger than the first pressure threshold P_{th1} and the length T_{ia} of the pause period stored in the operation storage unit **94** is larger than the predetermined pause threshold T_{thia} is satisfied, the start control is performed (see **S115**: No, **S117**: NO in FIG. 4, **S125**: Yes in FIGS. 5, and **S140**).

[0116] In step **118**, the control unit **92** switches the start control from the normal start control (see **S140** in FIG. 4). Specifically, the processing in step **119** and subsequent steps is performed.

[0117] In step **119**, the control unit **92** causes the adjusting unit **56** to supply the fuel gas. At this time, the main stop valve **54** and the pressure reducing valve **55** are closed. The process of step **119** is different from the process of step **122** of FIG. 3 in that the main stop valve **54** is closed.

[0118] The control unit **92** acquires the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** from the pressure sensor **57** when a predetermined time-threshold T_{th} has elapsed after the fuel gas is supplied to the adjusting unit **56** (see the right part of the middle stage in FIG. 1).

[0119] The processing of step **127** in FIG. 4 is the same as the processing of step **127** in the second embodiment.

[0120] In the anode gas supply passage **52**, when the pressure reducing valve **55** is capable of sufficiently shutting off the anode gas, when the fuel gas is supplied to the adjusting unit **56**, the anode gas in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** flows downstream. Therefore, the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** is greatly reduced. If so, a Yes determination is made at step **127**.

[0121] On the other hand, in the anode gas supply passage **52**, when the pressure reducing valve **55** cannot sufficiently shut off the anode gas, the anode gas between the pressure reducing valve **55** and the adjusting unit **56** flows downstream, while the high-pressure anode gas in the anode gas supply passage **52** between the pressure reducing valve **55** and the main stop valve **54** enters the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** via the pressure reducing valve **55**. Therefore, even if the fuel gas is supplied to the adjusting unit **56**, the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** does not significantly decrease within the time-threshold T_{th} . If so, a No determination is made at step **127**.

[0122] In the third embodiment, the main stop valve **54** is closed, and a drop in the pressure downstream of the pressure reducing valve **55** is confirmed (see **S119** in FIG. 4). Therefore, in a case where the pressure reducing valve **55** has failed and the flow of the gas cannot be sufficiently shut off, it is possible to reduce the possibility that the downstream adjusting unit **56** is destroyed

by being exposed to the high-pressure fuel gas, or the relief mechanism included in the adjusting unit **56** is opened and the fuel gas is ejected to the outside. For example, the relief mechanism may be opened as described above in a case where the valve element included in the adjusting unit **56** is configured to be pressed from the upstream side to the downstream side when the valve is closed. [0123] In step **127** of FIG. **4**, if the pressure P_m is less than the second pressure threshold P_{th2} , the process proceeds to step **130**. If the pressure P_m is greater than or equal to the second pressure threshold P_{th2} , the process proceeds to step **150**.

[0124] In a third embodiment, [0125] (i) The pressure P_m acquired by the pressure sensor **57** is larger than the first pressure threshold P_{th1} (see **S115**: No of FIG. **4**), [0126] (ii) The interruption of supplying power to the operation storage unit **94** is stored (see **S117**: Yes of FIG. **4**), and [0127] (iii) After the fuel gas is supplied to the adjusting unit **56** toward the fuel cell **20**, the pressure P_m acquired by the pressure sensor **57** becomes smaller than the predetermined second pressure threshold P_{th2} (see **S119** and **S127**: Yes in FIG. **4**), [0128] If the third start condition is satisfied, including (**S140**), start control is performed.

[0129] By performing such a process, it is possible to avoid a situation in which the start control is not performed even though there is no abnormality in the function of the pressure reducing valve **55** in the following cases. That is, when the length T_{ia} of the pause period in which the operation of the fuel cell **20** was not performed is larger than the pause threshold T_{thia} (refer to **S125**: Yes of FIG. **5**), the pressure in the anode gas supply passage **52** on the fuel cell **20** side is larger than the first pressure threshold P_{th1} with respect to the pressure reducing valve **55** (refer to **S115**: Yes of FIG. **4**), and the length T_{ia} of the pause period is not stored in the operation storage unit **94** because the supply of electric power to the operation storage unit **94** is interrupted (refer to **S117**: Yes of FIG. **4**), it is possible to avoid a situation in which the starting control is not performed. At this time, since it is conditioned that the pressure P_m acquired by the pressure sensor **57** becomes smaller than the second pressure threshold P_{th2} after the fuel gas is supplied to the adjusting unit **56** toward the fuel cell **20** (see **S127**: Yes in FIG. **4**), it is possible to avoid a situation in which the starting control is performed even though the function of the pressure reducing valve **55** is abnormal.

D. Other Embodiments

D1. Another Embodiment 1

[0130] (1) In the above-described embodiment, the adjusting unit **56** is a linear solenoid valve (see the middle portion of FIG. **1**). However, the adjusting unit **56** may have another configuration such as an injector.

[0131] (2) In the first embodiment, the pressure sensor **57** acquires the pressure in the anode gas supply passage **52** between the pressure reducing valve **55** and the adjusting unit **56** (see the right part of the middle stage in FIG. **1**). However, the pressure sensor **57** may acquire the pressure in the anode gas supply passage **52** between the adjusting unit **56** and the fuel cell **20** (see the right part of the middle stage in FIG. **1**). The processing of steps **110**, **115**, **119**, **127** may be performed based on measurements of the pressure sensor **58**.

[0132] (3) In the first embodiment, the first starting condition is that the pressure P_m acquired by the pressure sensor **57** is smaller than a predetermined first pressure threshold P_{th1} (see **S115**: Yes of FIG. **2**). However, the first start condition may include other weighting conditions such as, for example, that the operation mode of the fuel cell system **100** is set to a predetermined mode. The same applies to the second start condition and the third start condition.

[0133] (4) The second pressure threshold P_{th2} in the second embodiment and the third embodiment may be higher than the first pressure threshold P_{th1} , lower than the first pressure threshold P_{th1} , or equal to the first pressure threshold P_{th1} . The second pressure thresholds P_{th2} can be determined according to the hardware configuration of the anode gas supply passage **52**, the pressure reducing valve **55**, and the adjusting unit **56**, and the duration from the supply of the fuel gas to the adjusting unit **56** until the pressure sensor **57** measures the pressure. In the case where the second pressure

threshold value is set to be high, the detection of the decrease in pressure is performed when a short time elapses after the fuel gas is supplied to the adjusting unit **56**. When the second pressure threshold value is set lower than that, the detection of the decrease in pressure is performed when a longer time has elapsed since the fuel gas is supplied to the adjusting unit **56**.

[0134] (5) The second pressure threshold P_{th2} in the third embodiment is the same as the second pressure threshold P_{th2} in the second embodiment. However, when a decrease in pressure is detected in a state where the main stop valve **54** is closed, the second pressure threshold P_{th2} may be set lower than when a decrease in pressure is detected in a state where the main stop valve **54** is opened.

[0135] (6) In the above-described embodiment, in step **154**, the control unit **92** performs start control at the time of abnormality. Specifically, the control unit **92** performs setting so that, even if an instruction I_s indicating that the operation of the fuel cell **20** is to be started is input via the input unit **97**, the power is turned off immediately after the power is turned on (see **S150** of FIG. 2). However, in step **150**, the control unit **92** may only perform a process of closing the main stop valve **54** and stopping the fuel cell system **100** for the current instruction I_s indicating that the operation of the fuel cell **20** should be started without performing such a process.

[0136] (7) In the above-described embodiment, the fuel cell system **100** is mounted on a vehicle. However, the fuel cell system **100** may be mounted on another moving body such as a ship or an airplane, or may be installed in a building or the like.

D2. Another Embodiment 2

[0137] In the third embodiment, [0138] (i) The pressure P_m acquired by the pressure sensor **57** is larger than the first pressure threshold P_{th1} (see **S115**: No of FIG. 4), [0139] (ii) The interruption of supplying power to the operation storage unit **94** is stored (see **S117**: Yes of FIG. 4), and [0140] (iii) After the fuel gas is supplied to the adjusting unit **56** toward the fuel cell **20**, the pressure P_m acquired by the pressure sensor **57** becomes smaller than the predetermined second pressure threshold P_{th2} (see **S119** and **S127**: Yes in FIG. 4), [0141] If the third start condition is satisfied, including (**S140**), start control is performed.

[0142] However, as shown in the first embodiment, not the third start condition, the pressure P_m acquired by the pressure sensor **57** is larger than the first pressure threshold P_{th1} , and the length T_{ia} of the pause period stored in the operation storage unit **94** is larger than the predetermined pause threshold T_{thia} even when the second start condition including, is satisfied, start control may be performed (see **S115**: No, **S125**: Yes and **S140** in FIG. 2).

D3. Another Embodiment 3

[0143] After the main stop valve **54** is closed and the anode gas is supplied to the adjusting unit **56** toward the fuel cell **20**, the start control is performed when the third start condition including that the pressure P_m acquired by the pressure sensor **57** is smaller than the second pressure threshold P_{th2} within a predetermined time threshold T_{th} is satisfied (see **S119**, **S127**: Yes and **S140** in FIG. 4).

[0144] However, as shown in the second embodiment, it may be determined whether or not the pressure P_m acquired by the pressure sensor **57** is smaller than the second pressure threshold P_{th2} with the main stop valve **54** closed (see **S122** and **S127** in FIG. 3).

D4. Other Embodiment 4

[0145] In the second embodiment and the third embodiment, the start control is performed (**S140**) when the start condition including that the pressure P_m acquired by the pressure sensor **57** is smaller than the predetermined second pressure threshold P_{th2} (see **S127**: Yes and **S127**: Yes in FIG. 3) is satisfied after the fuel gas is supplied to the adjusting unit **56** toward the fuel cell **20**.

[0146] However, instead of such a condition, as shown in the first embodiment, the start control may be performed when the start condition including that the length T_{ia} of the pause period stored in the operation storage unit **94** is larger than the predetermined pause threshold T_{thia} is satisfied (see **S115**: No, **S125**: Yes and **S140** in FIG. 2).

[0147] The present disclosure is not limited to the embodiments above, and can be implemented with various configurations without departing from the scope of the present disclosure. For example, the technical features of the embodiments corresponding to the technical features in each mode described in the section of the summary of the disclosure may be replaced or combined appropriately to solve some or all of the above issues or to achieve some or all of the above effects. When the technical features are not described as essential in this specification, the technical features can be deleted as appropriate.

Claims

1. A fuel cell system comprising: a fuel cell; a fuel gas tank that holds a fuel gas; a gas flow path that connects the fuel cell and the fuel gas tank to allow the fuel gas to flow through the gas flow path; a pressure reducing valve provided in the gas flow path, capable of allowing the fuel gas supplied from the fuel gas tank to flow through the pressure reducing valve with a pressure of the fuel gas reduced, and capable of shutting off the fuel gas supplied from the fuel gas tank; a pressure sensor capable of acquiring a pressure in the gas flow path on a side of the fuel cell with respect to the pressure reducing valve; an operation storage unit that stores a length of a quiescent period of the fuel cell system; and a control unit that controls the fuel cell system, wherein: the control unit performs start control for starting operation of the fuel cell when an instruction to start the operation of the fuel cell is received, and when a first start condition is met, the first start condition including the pressure acquired by the pressure sensor being less than a first pressure threshold determined in advance; and the control unit performs the start control when a second start condition is met, the second start condition including the pressure acquired by the pressure sensor being more than the first pressure threshold and the length of the quiescent period stored in the operation storage unit being more than a quiescent threshold determined in advance.
2. The fuel cell system according to claim 1, further comprising: an adjusting unit provided in the gas flow path on a side of the fuel cell with respect to the pressure reducing valve to receive the fuel gas and supply the fuel gas to the fuel cell at a specified opening degree; and a power supply storage unit that stores a shutoff of supply of power to the operation storage unit, wherein: the pressure sensor acquires a pressure in the gas flow path between the pressure reducing valve and the adjusting unit; and the control unit performs the start control when an instruction to start the operation of the fuel cell is received and when a third start condition is met, the third start condition including (i) the pressure acquired by the pressure sensor being more than the first pressure threshold, (ii) a shutoff of supply of power to the operation storage unit being stored, and (iii) the pressure acquired by the pressure sensor becoming less than a second pressure threshold determined in advance after the adjusting unit is caused to supply the fuel gas toward the fuel cell.
3. The fuel cell system according to claim 2, further comprising a main stop valve provided in the gas flow path on a side of the fuel gas tank with respect to the pressure reducing valve to allow or shut off a flow of the fuel gas, wherein the pressure acquired by the pressure sensor becoming less than the second pressure threshold after the adjusting unit is caused to supply the fuel gas toward the fuel cell includes the pressure acquired by the pressure sensor becoming less than the second pressure threshold within a time threshold determined in advance after the adjusting unit is caused to supply the fuel gas toward the fuel cell with the main stop valve in a closed state.
4. A fuel cell system comprising: a fuel cell; a fuel gas tank that holds a fuel gas; a gas flow path that connects the fuel cell and the fuel gas tank to allow the fuel gas to flow through the gas flow path; a pressure reducing valve provided in the gas flow path, capable of allowing the fuel gas supplied from the fuel gas tank to flow through the pressure reducing valve with a pressure of the fuel gas reduced, and capable of shutting off the fuel gas supplied from the fuel gas tank; an adjusting unit provided in the gas flow path on a side of the fuel cell with respect to the pressure reducing valve to receive the fuel gas and supply the fuel gas to the fuel cell at a specified opening

degree; a pressure sensor capable of acquiring a pressure in the gas flow path between the pressure reducing valve and the adjusting unit; and a control unit that controls the fuel cell system, wherein: the control unit performs start control for starting operation of the fuel cell when an instruction to start the operation of the fuel cell is received, and when a start condition is met, the start condition including the pressure acquired by the pressure sensor being less than a first pressure threshold determined in advance; and the control unit performs the start control when an additional start condition is met, the additional start condition including (i) the pressure acquired by the pressure sensor being more than the first pressure threshold, and (ii) the pressure acquired by the pressure sensor becoming less than a second pressure threshold determined in advance after the adjusting unit is caused to supply the fuel gas toward the fuel cell.
